TwinWorld 2026 Challenge Final Verification Factsheet Offset Sheets and Scorer-Matched Resolution for Large-Scale Semantic Gaussian Splatting Reconstruction

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1. Introduction

This factsheet describes the solution submitted by team `Manish` to the TwinWorld 2026 Challenge on large-scale 3D semantic Gaussian Splatting reconstruction. The final submission placed 5th of 21 entries with a final score of 1.4837, and obtained the highest geometry F-score of any entry in the challenge (0.6386).

The pipeline is deliberately conventional in its components – COLMAP multi-view stereo, gsplat 3D Gaussian Splatting, and gradient-boosted trees over local geometric descriptors – and the contribution lies in two measurements about how the three scored terms respond to submission-side decisions. The first is that the semantic term, which is a nearest-neighbour label transfer within a fixed radius and carries no precision penalty, is limited by how densely space near the reconstructed surface is populated rather than by label quality. The second is that the submission voxel is not a free hyperparameter but is dictated by the evaluation protocol itself.

2. Team details

- **Team name:** `Manish`
- **Team leader:** Manish Joshi, Independent Researcher, `manishinuae@gmail.com`
- **Other team members:** none (single-participant team)
- **Team website:** n/a
- **Codabench user name:** `manish`
- **Codabench ID of the final submission:** `910133` (Final phase, submitted 2026-08-31 15:45 UTC)
- **Final challenge score:** 1.4837 (see Tab. 1)
- **Source code, pretrained models and checkpoints:** `https://huggingface.co/manishjoshiai/twinworld2026-verification` files `twinworld-code.zip` and `twinworld-semantic-checkpoints.tar`
- **Complete results corresponding to the final submission:** same repository, `reconstruction/` (thirteen per-scene archives of the native Gaussian representation) and `submission/` (the submitted point clouds and renders)

The repository is private; a read-only access token scoped to it accompanies this factsheet in the submission e-mail.

Metric	Value
Final score	1.4837
PSNR	20.90
SSIM	0.7432
LPIPS	0.2996
rendering	*0.6205*
F-score (TUM)	**0.6386**
mIoU 3D (Gold Coast)	0.2247

Table 1. Final-phase scores of Codabench submission 910133. The F-score is the highest of all 21 entries in the challenge.

3. Method details

3.1. Overall pipeline

Figure 1 summarises the pipeline.

The pipeline has three branches, one per scored term, sharing a common structure-from-motion and multi-view stereo front end.

1. **Front end.** The provided COLMAP poses are used as given. Training images are upsampled $2\times$ with Lanczos resampling before undistortion, with intrinsics scaled to match, and dense reconstruction is run with a permissive photometric-consistency setting. Per-scene depth maps are fused into a point cloud, and a union of fusion passes is retained per scene.
2. **Rendering branch.** 3D Gaussian Splatting is optimised per scene with gsplat and the test poses are rendered.

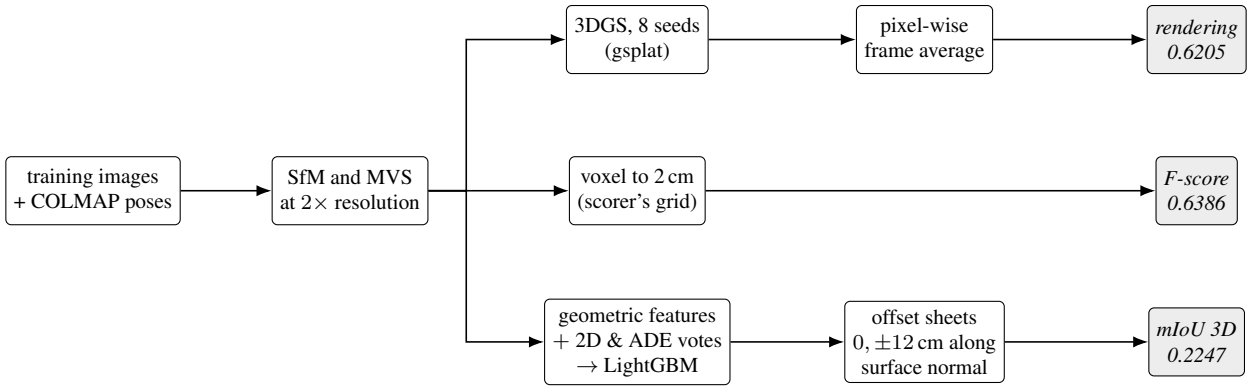


Figure 1. Pipeline. One shared reconstruction front end feeds three independent branches, one per scored term. The two contributions are on the right-hand side of the lower branches: submitting TUM at the scorer's own 2 cm grid rather than a coarser one, and thickening the labelled Gold Coast cloud perpendicular to its surface so that ground truth displaced from our reconstruction still falls within the 10 cm match radius. Values shown are the final-phase scores of submission 910133.

Eight independently seeded models are trained per scene and their rendered frames are averaged pixel-wise.

3. **Geometry branch (TUM).** The fused MVS cloud is voxel-downsampled to 2 cm and submitted directly.
4. **Semantic branch (Gold Coast).** Per-point features are classified into the five challenge classes, the labelled cloud is thickened perpendicular to its own surface, and the result is submitted with per-point labels.

3.2. Rendering

Gaussian models are optimised with gsplat using spherical harmonics of degree 3, the default densification strategy, antialiased rasterisation, and a D-SSIM weight of 0.2. Two per-dataset settings were selected on the development scenes: Gold Coast uses the sparse SfM points as initialisation and 7000 iterations, while TUM initialises from the dense MVS cloud and uses 3000 iterations. Densification stops at 72% of the schedule.

Repeating an identical configuration on a Gold Coast development scene produced a rendering score between 0.5103 and 0.5386 across three runs – a standard deviation of 0.014, against 0.0018 on TUM. The held-out Gold Coast views extrapolate 16–18 m from the nearest training camera, so small differences in densification move the rendered image substantially. Averaging eight independently seeded runs is therefore variance reduction, and it improves all three rendering metrics simultaneously, including LPIPS, which frame averaging normally harms: the floaters each run invents independently cancel, while the geometry the runs agree on does not. Nothing about the held-out image is used to combine the runs.

3.3. Geometry and the submission voxel

Image resolution, not MVS parameter tuning, was the binding constraint on TUM geometry. The ground sampling distance is about 3.4 cm/px at 36 m altitude while the ground truth is a 2 cm scan. Upsampling the input images 2× improved mean F from 0.4758 to 0.5336 on a development scene, with F@5 improving by 47%; by comparison, an entire sweep over MVS filtering settings moved the mean by 0.002. A 4× variant was also reconstructed and measured, and it *lost* to the retained fusion union (0.5539 against 0.5905), so it was not adopted.

The submission voxel proved equally consequential and is a property of the evaluation protocol rather than a tunable. The scorer voxel-downsamples both the submitted and reference clouds to 2 cm before matching, so a submission coarser than 2 cm discards resolution the scorer would have used, while a submission finer than 2 cm contributes points that are immediately merged. Measured on a development scene:

Submission voxel	Points	Mean F
4 cm	5.17 M	0.5698
2 cm	8.37 M	**0.5905**

Table 2. TUM geometry against the submission voxel, development scene.

3.4. Semantic labelling

Labels are produced by a LightGBM classifier over a concatenation of three feature groups, fitted on the two labelled Gold Coast scenes and applied to the withheld ones.

Geometric features. Local structure-tensor descriptors – linearity, planarity, sphericity, verticality, log local extent, and signed offset from the locally fitted plane – computed at several neighbourhood sizes, together with height above a local terrain model and signed height relative to the local roof plane. Neighbourhoods are k-nearest rather than

radius-based so the descriptor is scale-adaptive.

Learned 2D votes. The provided 3D semantic ground truth is projected into the training views to synthesise 2D masks, which requires no additional annotation. A SegFormer `mit-b0` backbone is fine-tuned on those masks and its per-view predictions are lifted back to 3D by projecting every point into every view and accumulating a vote histogram. A depth buffer over our own cloud suppresses votes collected by occluded points. This addresses a failure that geometry cannot: a window is coplanar with its wall, so plane offset, verticality and curvature are identical for both, while in image space windows are 18% of labelled pixels.

Zero-shot votes. A SegFormer-B2 model fine-tuned on ADE20K is applied without adaptation and its 150 classes are mapped into eight super-groups (building, window, road/ground, vegetation, vehicle, sky, object, water). These votes are projected with the same occlusion-aware procedure. This recovers label-to-class conversion in regions where 23 training images per scene are insufficient.

Predictions are made on a thinned cloud and propagated to the full submission cloud by $k{=}5$ nearest-neighbour majority vote.

3.5. Offset sheets

The semantic metric assigns each ground-truth point the label of the nearest submitted point within 10 cm and counts unmatched points as errors. It applies no penalty for submitted points that are wrong or superfluous. Our reconstructed surface sits a median 13.8 cm from the ground truth, so a majority of the ground truth was unmatchable irrespective of label quality, and the measured coverage as a function of match radius (0.42 at 10 cm, 0.71 at 20 cm, 0.90 at 50 cm) shows the mass is present but displaced.

Densifying *along* the surface cannot address this, because additional points land on the same displaced surface. The submitted cloud is therefore thickened *perpendicular* to it: each point emits copies at ± 12 cm along its local surface normal, carrying its parent's label. Normals are fitted by local PCA on the clean 10 cm surface cloud and oriented towards the camera rig, then inherited by the packed cloud.

Configurations were ranked by an analytic estimator. Because the metric depends only on which point wins the nearest-neighbour race, the distance from a ground-truth point g to the sheet of a parent p with unit normal n at offset o is

$$\|d - o\,n\|^2 = \|d\|^2 - 2\,o\,(d{\cdot}n) + o^2, \qquad d = g - p,$$

so a single k-nearest-neighbour query of the ground truth against the parents prices every offset configuration in closed form without materialising any sheet. The estimator reproduced the reference scorer exactly on both labelled scenes.

Three findings determined the shipped configuration. The gain must be bought with additional points and cannot be reallocated: every budget-matched variant, in which parents were coarsened to pay for the sheets, was neutral or worse, and coarsening past roughly 15 cm breaks lateral coverage because the worst-case in-plane gap then exceeds the 10 cm match radius. Reach matters more than sheet count, up to a saturation near ± 18 cm. And the effect reverses at long range: a stack reaching ± 54 cm achieved the highest coverage measured anywhere in our experiments and the *lowest* mIoU of any sheeted configuration, because a copy displaced half a metre carries a label sampled too far away to be informative and, being nearer to the ground-truth point than the honest surface behind it, displaces a label that would have been correct.

3.6. External data and models

Do you use extra data besides the competition data provided? No. All reconstruction, all classifier training and all Gaussian optimisation used only the challenge-provided images, poses and labels. No additional imagery, point clouds or annotations of any kind were introduced. The 2D masks used to supervise the segmenter were not annotated: they are synthesised by projecting the challenge's own 3D semantic ground truth into the training views.

Two publicly available pretrained *models* were used, and are disclosed here in full:

- `nvidia/mit-b0` – SegFormer [3] backbone with ImageNet pretraining, fine-tuned on 2D masks synthesised by projecting the challenge's own 3D labels. No external segmentation annotation was used.
- `nvidia/segformer-b2-finetuned-ade-512-512` – SegFormer fine-tuned on ADE20K [5], applied zero-shot with its output classes mapped to eight super-groups.

Third-party software: COLMAP [2] for SfM and MVS, gsplat [4] for Gaussian Splatting, LightGBM [1] for classification.

3.7. Novelty and prior publication

None of this work has been previously published, and no part of it is under review elsewhere. It was developed entirely within the challenge period.

The components are all established: COLMAP for structure-from-motion and multi-view stereo, gsplat for Gaussian Splatting, SegFormer for 2D segmentation, and gradient-boosted trees for point classification. We claim no novelty for any of them, and we would not describe the solution as a new method.

What we would put forward as contributions are two measurements about the evaluation protocol, both of which we believe are of interest beyond our own submission. The first is that the semantic metric – a nearest-neighbour label transfer within a fixed radius, with no precision penalty – is

bounded by how densely space near the reconstructed surface is populated rather than by label quality, and that thickening the cloud perpendicular to its surface exploits this, but saturates and then reverses once the displacement exceeds roughly a quarter of a metre because coverage and label conversion are not independent. The second is that the submission voxel is dictated by the scorer's own downsampling grid rather than being a free hyperparameter, which we established only on the final day of the challenge and which was worth more per unit of effort than anything else we attempted.

The offset-sheet operation itself is, as far as we are aware, not standard practice; whether it should be is a separate question, and Sec. 5 argues that a symmetric metric would remove the incentive for it.

3.8. Complexity and runtime

Per Gold Coast scene, a single Gaussian model converges to roughly 0.9 M Gaussians in about 140 s on one NVIDIA A100-80GB; the eight-member ensemble costs about 19 minutes per scene. MVS at $2\times$ resolution is the dominant cost at roughly 700–1700 s per scene. The semantic classifier is a 200-estimator LightGBM model and trains in minutes on CPU. Sheet materialisation and packaging are CPU-only and take under 10 minutes for the whole submission. Peak GPU memory during Gaussian optimisation is under 40 GB.

4. Ensemble and Fusion Strategies

Two fusion strategies are used, at different stages.

Rendering ensemble. Eight independently seeded Gaussian models per scene, averaged pixel-wise in image space. This was adopted because per-run variance on Gold Coast is an order of magnitude larger than on TUM, and averaging beats not only the mean single run but the best single run while improving PSNR, SSIM and LPIPS simultaneously. An MCMC densification variant was measured as an alternative and rejected: it improved PSNR markedly (21.99 against 19.94) but degraded LPIPS enough (0.457 against 0.418) to lose overall, 0.5650 against 0.5759.

Semantic feature fusion. The classifier consumes geometric descriptors, supervised 2D votes and zero-shot ADE20K votes as a single concatenated feature vector rather than combining three separate predictions. Each group contributes where the others fail: geometry cannot express a window, the supervised segmenter is limited by 23 images per scene, and the zero-shot model has no notion of the challenge's class definitions but is reliable on building, vegetation and ground. Adding the zero-shot group to the previously adopted stack improved leave-one-scene-out mIoU by 0.0047.

Every candidate throughout the challenge was admitted through a fixed gate: both leave-one-scene-out folds had to improve over their own baselines and the pooled score had to beat the incumbent. Roughly 90 configurations were evaluated and the large majority were rejected under this rule, including nine consecutive attempts to improve semantics by increasing model capacity or training schedule.

5. Other Details

Impressions. The challenge is well designed in that the three scored terms genuinely pull in different directions, and the Gold Coast semantic task is a real research problem rather than a tuning exercise: two labelled scenes are not enough supervision for the withheld ones, and our nine rejected capacity experiments reflect that rather than any shortage of effort.

Difficulties. The dominant difficulty was that the submission envelope was never documented. An early submission failed at 108.6 M points and we inferred a size limit from it, then spent ten days optimising under a point budget that did not exist – the Final phase in fact specifies no size limit at all, only a 6000 s execution-time cap, and the figure we had inferred was almost certainly the Development phase's tighter time limit. Recognising this on the final day and restoring TUM to the scorer's own 2 cm grid was worth 0.011 of final score for one measurement, more per unit of effort than anything else attempted in the last week of the challenge.

Comments on the evaluation. The semantic metric's asymmetry – a nearest-neighbour transfer with no precision penalty – rewards filling space near the surface, which is a submission-side strategy rather than a reconstruction improvement. Our own results illustrate both the exploit and its limit: sheets are worth roughly $+0.042$ of final score, but the effect saturates and then reverses, because coverage and label conversion are not independent. A symmetric metric, or a cap on submitted points relative to the reference cloud, would remove the incentive without weakening the task. Publishing the size and runtime limits explicitly would also help; we are unlikely to have been the only team to misread a failed submission as a size limit.

Suggestions. A semantic sub-track evaluated at several match radii would separate reconstruction accuracy from space filling, and would have made visible much earlier that the leading teams' advantage was that their surfaces fall inside the 10 cm radius to begin with, rather than that they submit more points.

References

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